CBCS SCHEME

USN

BCS405A

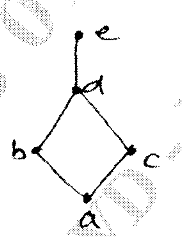
Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025

Discrete Mathematical Structures

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Tautology, show that $[(p\vee q)\wedge\{(p\to r)\wedge(q\to r)\}]\to r$	6	L1	CO1
	b.	Prove the following using the laws of logic : $\neg[\{(p\vee q)\wedge r\}\to\neg q]\Leftrightarrow\neg[\neg[(p\vee q)\wedge r]\vee.\neg q]\Leftrightarrow q\wedge r$.	7	L2	CO1
	c.	Give i) a direct proof ii) an Indirect proof for the following statement "If n is an odd integer then n + 9 is an even integer".	7	L2	CO1
		OR			
Q.2	a.	Define i) an open statement ii) quantifiers.	6	L2	CO1
	b.	Test the validity of the following arguments. i) $\begin{array}{l} p\wedge q \\ \underline{p\to(q\to r)} \\ \therefore r\end{array}$ ii) $\begin{array}{l} P \\ P\to\sim q \\ \underline{\sim q\to\sim r} \\ \therefore\sim r\end{array}$	7	L2	CO1
	c.	For the following statements the universe comprises all non – zero integers. Determine the truth value of each statement. i) $\exists x, \exists y\ [xy=1]$ ii) $\exists x, \forall y\ [xy=1]$ iii) $\forall x, \exists y\ [xy=1]$ iv) $\exists x, \exists y\ [(2x+y=5)\wedge(x-3y=-8)]$ v) $\exists x, \exists y\ [(3x-y=17)\wedge(2x+4y=3)]$.	7	L2	CO1
		Module – 2			
Q.3	a.	Define the well ordering principle. By Mathematical induction, prove that $1+2+3+\ldots\ldots+n=\frac{1}{2}n(n+1)$, $n\in z^{+}$.	6	L2	CO2
	b.	Prove that $F_n=\frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n-\left(\frac{1-\sqrt{5}}{2}\right)^n\right]$. For F_0, F_1, F_2, ... are the Fibonacci numbers.	7	L2	CO2
	c.	Find the number of permutations of the letters of the word 'MASSASAUGA'. In how many of these all four A's are together? How many of them begin with S's?	7	L3	CO2
		OR			

Q.4	a.	Prove that $4n < n^2 - 7$ for all positive integers $n \geq 6$.	6	L2	CO3
	b.	Find the co-efficients of $x^9 y^3$ in the expansion of $(2x - 3y)^{12}$.	7	L3	CO3
	c.	Let $a_0 = 1$, $a_1 = 2$, $a_3 = 3$ and $a_n = a_{n-1} + a_{n-3}$ for $n \geq 3$, prove that $a_n \leq 3^n$ for all +ve integers n.	7	L2	CO3
Module – 3					
Q.5	a.	State Pigeon hole principle. Prove that if 30 dictionaries in a library contains a total of 61,327 pages then atleast one of dictionaries must have atleast 2045 pages.	6	L2	CO3
	b.	Define power set. For any sets A, B, $C \leq U$, prove that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.	7	L2	CO3
	c.	Let f and g be functions from R to R defined by $f(x) = ax + b$ and $g(x) = 1 - x + x^2$ if $(gof)(x) = 9x^2 - 9x + 3$, determine a & b.	7	L3	CO3
OR					
Q.6	a.	Let $f : R \rightarrow R$ be defined by $f(x) = \begin{cases} 3x - 5, \text{if } x > 0 \\ 1 - 3x, \text{if } x \leq 0 \end{cases}$ Find $f^1(-5, 5)$ and $f^1(-6, 5)$.	6	L2	CO3
	b.	Let N be the set of Natural numbers. Let a relation R be defined by $R = \{(a, b) / a \in N , b \in N , a - b \text{ is divisible by } 5\}$. Prove that R is an equivalence relation.	7	L2	CO3
	c.	For A = { a, b, c, d, e}, the Hasse diagram for the poset(A, R) is as shown below : i) Determine the relation matrix for R ii) Construct the diagraph for R. 	7	L3	CO3
Module – 4					
Q.7	a.	Determine the number of integers between 1 and 250 that are divisible by 3 and not divisible by 5 and 7.	6	L3	CO4
	b.	Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$, where $n \geq 0$ and $F_0 = 0$, $F_1 = 1$.	7	L2	CO4
	c.	Define Derangement. Find the number of derangement of 1, 2, 3, and 4.	7	L3	CO4
OR					

Q.8	**a.**	Find the Rook polynomial for the chess board contain 4 squares as shown in the Fig.Q8(a). $\begin{array}{\|c\|c\|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array}$ Fig.Q8(a)	**6**	**L3**	**CO4**
	b.	Solve the recurrence relation $a_n = 5a_{n-1} + 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 3$.	**7**	**L2**	**CO4**
	c.	Find the distinct numbers which are multiples of at least one of 15, 40 and 35 not exceeding 1000.	**7**	**L3**	**CO4**
		Module – 5			
Q.9	**a.**	Define group and subgroup with example each.	**6**	**L1**	**CO5**
	b.	State and prove Lagrange's theorem.	**7**	**L2**	**CO5**
	c.	Define Klein 4 group. Verify A = {e, a, b, c} is a Klein 4 group.	**7**	**L2**	**CO5**
		OR			
Q.10	**a.**	Prove that the intersection of two subgroup of a group is a subgroup of the group.	**6**	**L2**	**CO5**
	b.	Prove that the cube roots of unity form a group under the multiplication.	**7**	**L2**	**CO5**
	c.	Let G = S_4, the symmetric group of order 4, for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$, find the subgroup H = < a >, determine the number of left cosets of H in G.	**7**	**L3**	**CO5**

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Discrete Mathematical Structures (BCS405A)

Scheme of Solution.

Q1.a. A compound proposition which is always true regardless of truth of its component is called tautology.

$[(p \vee q) \wedge \{(p \to r) \wedge (q \to r)\}] \longrightarrow r.$

p	q	r	$p \vee q$ (x)	$p \to r$ (y)	$q \to r$ (z)	$x \wedge y \wedge z$	$(x \wedge y \wedge z) \to r$
0	0	0	0	1	1	0	1
0	0	1	0	1	1	0	1
0	1	0	1	1	0	0	1
0	1	1	1	1	1	1	1
1	0	0	1	0	1	0	1
1	0	1	1	0	1	0	1
1	1	0	1	1	0	0	1
1	1	1	1	1	1	1	1

b. $\sim[\{(p \vee q) \wedge r\} \to \sim q] \Leftrightarrow \sim[\sim[(p \vee q) \wedge r] \vee \sim q] \Leftrightarrow q \wedge r$

$a = (p \vee q) \wedge r$, $b = \sim q$.

$\therefore$ $a \to b \equiv \sim a \vee b$

$\therefore \{(p \vee q) \wedge r\} \to \sim q \equiv \sim\{(p \vee q) \wedge r\} \vee \sim q$ (Implication law)

$\sim[\sim\{(p \vee q) \wedge r\} \vee \sim q] \equiv \sim[\sim\{(p \vee q) \wedge r) \wedge q\}]$ (De Morgan law)

$\because \sim P \vee \sim Q \equiv \sim(P \wedge Q)$.

$\equiv [((p \vee q) \wedge r) \wedge q]$ (law of double negation)

$\equiv (p \vee q) \wedge (r \wedge q)$ (Associative law)

$\equiv (p \vee q) \wedge (q \wedge r)$ (Communicative law).

$\equiv \{(p \vee q) \wedge q\} \wedge r$ (Associative law.

$\equiv q \wedge r$ (Absorption law).

c. i) Direct Proof: Assume that n is an odd integer. Then $n=2k+1$ for some integer k. This gives $n+9=(2k+1)+9$ $=2(k+5)$ for which ~~it~~ is evident that $n+9$ is even. This establishes the truth of statement is direct Proof.

ii) Indirect Proof: Assume that $n+9$ is not an even integer. Then $n+9=2k+1$ for some integer k. This gives $n=(2k+1)-9$ $=2(k-4)$ which shows that n is even. Thus if $n+9$ is not even, then n is not odd. This proves the Contrapositive of given Statement. This proof of the Contrapositive serves as an indirect proof of given statement.

2 a. i) An Open statement :- Consider the declarative Sentences, $\frac{x}{3}$, $x^2=2$, $x+5=8$ etc. which are not proposition unless the symbol x is specified Such Statements are called open statement or open sentences.

ii) quantifiers :- The words 'all', 'every', 'some', 'any', 'there exists' etc which are associated with open statements with the idea of a quantity are called quantifiers.

b. i)

$$\begin{array}{l} p \wedge q \\ p \to (q \to r) \\ \hline \therefore r \end{array}$$

Step 1: Assume the premises are true. If $(p \wedge q)$ is true, then both p & q must be true individually.

Step 2: $p \to (q \to r)$, we know that p is true, the consequent $(q \to r)$ must be true.

Step 3: $(q \to r)$ is true and from step 1 that q is true.

Step 4: For the implication $(q \to r)$ to remain true while q is true, r must be true.

Since r is to be true whenever $p\wedge q$ & $\{p\rightarrow(q\rightarrow r)\}$ are true, the argument is valid.

ii) p

$p\rightarrow\sim q$

$\sim q\rightarrow\sim r$

$\therefore \sim r$.

Step 1: Assume the p, $p\rightarrow\sim q$ & $\sim q\rightarrow\sim r$ are true. We start with fact that p is true.

Step 2: p to $(p\rightarrow\sim q)$, this forces $\sim q$ to be true.

Step 3: $\sim q$ to $(\sim q\rightarrow\sim r)$ is true.

Step 4: Final conclusion $\sim r$, to be true.

c. i) For some $x=1$ and some $y=1$, we have $1\cdot1=xy=1$ is true. Therefore it's true.

ii) For some $x=2$, $y=3$, $xy=2\cdot3=6\neq1$ so is not true.

iii) For some $y=1$, $x=4$, $xy=4\cdot1=4\neq1$ 'cis not true.

iv) Solving $2x+y=5$, $x-3y=-8$ we get $x=1$, $y=3$, true then Both $x=1$ & $y=3$ is true.

v) Solving $3x-y=17$, $2x+4y=3$, since $\frac{31}{14}$ is not a integer, there is no solution within our universe. Hence $x=31/14$ & $y=-1/14$ is not true.

Q3.

a) Every non empty subset of Z^+ contains a smallest element called well ordering principle.

$1+2+3+\cdots+n=\frac{1}{2}n(n+1)$, $n\in Z^+$.

Soln: Let $S(n)$: $1+2+3+\cdots+n=\frac{1}{2}n(n+1)$

Basic step: $S(1)$ is the statement.

$1 = \frac{1}{2}(1)(1+1) = \frac{1}{2}(2)$ is true.

Thus the statement $S(n)$ is verified by $n=1$.

Induction step: Assume that $S(n)$ is true for $n=k$ where $k \geq 1$ i.e assume that $S(k): 1+2+3+\cdots+k = \frac{1}{2}k(k+1)$ is true.

adding $(k+1)$ to both sides, we get.

$$1+2+3+\cdots+k+k+1 = \frac{1}{2}k(k+1)+(k+1)$$
$$= \frac{1}{2}(k+1)(k+2).$$

This statement $S(k+1)$ is true whenever $S(k)$ is true for $k \geq 1$.

Hence M.I, $S(n)$ is true $\forall n \geq 1$.

b. Let $S(n)$: $F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$

Basic step $S(0)$: $F_0 = \frac{1}{\sqrt{5}}[1-1] = 0$

$$S(1): F_1 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^1 - \left(\frac{1-\sqrt{5}}{2}\right)\right]$$
$$= \frac{1}{\sqrt{5}}\left[\frac{1}{2}+\frac{\sqrt{5}}{2}-\frac{1}{2}+\frac{\sqrt{5}}{2}\right]$$
$$= \frac{1}{\cancel{\sqrt{5}}} \cdot \frac{\cancel{2}\cancel{\sqrt{5}}}{\cancel{2}} = 1.$$

Thus $S(n)$ is true for $n=0,1$.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$

i.e $S(k)$: $f_k = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^k - \left(\frac{1-\sqrt{5}}{2}\right)^k\right] \quad \forall k \geq 1$

Theroem prove that $S(k+1)$ is true.

i.e $F_{k+1} = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right]$

Consider $F_{k+1} = F_k + F_{k-1}$ $\quad (\because F_n = F_{n-1}+F_{n-2})$

$$F_{k+1}=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^k-\left(\frac{1-\sqrt5}{2}\right)^k\right]+\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{k-1}-\left(\frac{1-\sqrt5}{2}\right)^{k-1}\right]$$

$$=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{k-1}\left\{\frac{1+\sqrt5}{2}+1\right\}-\left(\frac{1-\sqrt5}{2}\right)^{k-1}\left\{\frac{1-\sqrt5}{2}+1\right\}\right]$$

$$=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{k-1}\left(\frac{3+\sqrt5}{2}\right)-\left(\frac{1-\sqrt5}{2}\right)^{k-1}\left(\frac{3-\sqrt5}{2}\right)\right]$$

$$=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{k-1}\left(\frac{1+\sqrt5}{2}\right)^2-\left(\frac{1-\sqrt5}{2}\right)^{k-1}\left(\frac{1-\sqrt5}{2}\right)^2\right]$$

$$\left[\because \left(\frac{1+\sqrt5}{2}\right)^2=\frac{1+5+2\sqrt5}{4}=\frac{6+2\sqrt5}{4}=\frac{\cancel{2}(3+\sqrt5)}{\cancel{4}\,2}=\frac{3+\sqrt5}{2}\right.$$

$$\therefore F_{k+1}=\frac{1}{\sqrt5}\left[\left(\frac{1+\sqrt5}{2}\right)^{k+1}-\left(\frac{1-\sqrt5}{2}\right)^{k+1}\right]$$

C. i) Given word has 10 letters of which 4 are A, 3 are S & 1 each are M, U, G.

$\therefore$ required no. of permutations is $\frac{10!}{4!\,3!\,1!\,1!\,1!}$

$= 25,200.$

ii) If in a permutation, all A's are to be together, we treat all of A's as one single letter say x. Then the letters to be permuted read (AAAA), S, S, S, M, U, G which are 7 in no.

$\therefore$ No. of permutations is $\frac{7!}{1!\,3!\,1!\,1!\,1!} = 840.$

iii) If in a permutation, beginning with S, there occur nine empty positions to fill, where two are S, 4 are A, 1 each are M, U, G

$\therefore$ No. of such permutations $= \frac{9!}{2!\,4!\,(1!)^3} = 7560.$

4a. Let $S(n) : 4n < (n^2 - 7)$

To prove that $S(n)$ is true $\forall$ +ve integers $n \geq n_0$ where $n_0 = 6$.

Basic step: $S(6) : (4 \times 6) < (6^2 - 7)$

i.e $24 < 29$ which is true.

$\therefore S(n)$ is true for $n = n_0 = 6$.

Induction Step: Assume that $S(n)$ is true for $n = k$

$k \geq 6$ i.e $4k < (k^2 - 7)$ for $k \geq 6$

add 4 on b.s.

then $4(k+1) = 4k + 4 < (k^2 - 7) + 4$

$< (k^2 - 7) + (2k + 1)$.

Because when $k \geq 6$ we have $2k \geq 12$

$\Rightarrow 2(k+1) \geq 13 > 4$

$\therefore 4(k+1) < (k^2 + 2k + 1) - 7$

i.e $4(k+1) < (k+1)^2 - 7$ This is the Statement $S(k+1)$

$\therefore S(k+1)$ is true whenever $S(k)$ is true $\forall$ $k \geq 6$

Thus $S(n)$ is true $\forall$ $n \geq 6$.

b. By Binomial theorem.

$$(2x - 3y)^{12} = \sum_{r=0}^{12} \binom{12}{r} (2x)^{12-r} (-3y)^r$$

$$= \sum_{r=0}^{12} \binom{12}{r} 2^{12-r} \cdot x^{12-r} \cdot (-3)^r \cdot y^r$$

Taking $r = 3$ in the above expansion, the coefficient of $x^9 y^3$ is

$$\binom{12}{3} (2x)^9 (-3y)^3 \Rightarrow \binom{12}{3} = \frac{12 \times 11 \times 10}{3 \times 2 \times 1} = 220.$$

$\therefore 2^9 = 512$ and $(-3)^3 = -27$.

$\therefore$ The coefficient of $x^9 y^3$ is $-3{,}041{,}280$.

c. Let $S(n) : a_n \leq 3^n$

Basic step : $S(0) = a_0 \leq 3^0$

$= 1 \leq 1$ true.

$S(1) = a_1 \leq 3^1$

$= 2 \leq 3$. True.

$S(2) = a_2 \leq 3^2$

$= 3 \leq 9$

Thus $S(n)$ is true for $n = 0, 1, 2$.

Induction step: Assume $S(K)$ is true.

i.e $S(K) : a_K \leq 3^K$. $\forall\ K \geq 3$.

Consider $a_n = a_{n-1} + a_{n-3}$. (using definition a_n)

$a_{n-1} \leq 3^{n-1}$ — (1)

$\therefore a_{n-3} \leq 3^{n-3}$ — 2.

Substitute (1) & (2)

$a_n \leq 3^{n-1} + 3^{n-3}$.

To prove $a_n \leq 3^n$ we Check $3^{n-1} + 3^{n-3} \leq 3^n$.

Factor out. 3^{n-3} . from right side $3^{n-3}(3^2+1)$

$3^{n-3}(9+1) = 10 \cdot 3^{n-3}$.

$10 \cdot 3^{n-3} \leq 3^n$.

Since $3^n = 3^3 \cdot 3^{n-3} = 27 \cdot 3^{n-3}$.

$\therefore 10 \cdot 3^{n-3} \leq 27 \cdot 3^{n-3}$

Thus. $a_n \leq 3^n$ is true.

Hence M.I required result is true $n \geq 3$.

5a. Pigeon hole Principle : If 'm' pigeons occupy 'n' pigeonhole & if $m>n$, then two or more pigeons occupy the same pigeonhole.

Consider 61,327 pages are pigeons i.e $m=61,327$ and 30 dictionaries as pigeon holes i.e $n=30$.

By using the generalised Pigeonhole principle, atleast 1 dictionary must contains $p+1$ or more ~~pigeon~~ pages.

$$\text{i.e } p+1 = \left\lceil \frac{m-1}{n} \right\rceil + 1$$

$$= \left\lceil \frac{61,327-1}{30} \right\rceil + 1$$

$$= 2044 + 1$$

$$= 2045.$$

This process the reqd result.

b. Power set denoted as $P(S)$ is the set of all possible subsets of S including the empty set ϕ and the set S itself.

$\Rightarrow A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$

Let $(x,y) \in A \times (B \cup C)$.

By defn of Cartesian product $x \in A$ & $y \in (B \cup C)$

By defn of Union, $y \in B$ or $y \in C$

$\therefore$ $x \in A$ & $y \in B$ then $(x,y) \in A \times B$.

$x \in A$ & $y \in C$ then $(x,y) \in A \times C$

$\therefore$ $(x,y) \in (A \times B) \cup (A \times C)$.

$\Rightarrow (A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$

Let $(x,y) \in (A \times B) \cup (A \times C)$

By defn of union $(x,y) \in (A \times B)$ or $(x,y) \in A \times C$.

Consider $(x,y) \in A \times B$ then $x \in A$ & $y \in B$.

$(x,y) \in A \times C$ then $x \in A$ & $y \in C$.

Both cases $x \in A$. Since $y \in B$ or $y \in C$ it follows.

$y \in (B \cup C)$.

By def^n of Cartesian product. $(x,y) \in A \times (B \cup C)$.

$\therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$.

c. $(g \circ f)(x)$ means Substitute $f(x)$ into function $g(x)$.

$$g(f(x)) = 1 - (ax+b) + (ax+b)^2$$

Expand the term.

$$-(ax+b) = -ax-b. \quad \text{---(1)}$$

$$(ax+b)^2 = a^2x^2+2abx+b^2 \quad \text{---(2)}$$

Combined (1) & (2)

$$(g \circ f)(x) = a^2x^2 + (2ab-a)x + (b^2-b+1)$$

Compare the expression $(g \circ f)(x) = 9x^2 - 9x + 3$.

for x^2 terms : $a^2 = 9$

x terms : $2ab - a = -9$

Constant terms : $b^2 - b + 1 = 3$.

Solve a & b.

$$a^2 = 9 \Rightarrow a = 3 \text{ or } a = -3.$$

If $a=3$. into x coefficient.

$$2(3)b - 3 = -9$$

$$6b - 3 = -9$$

$$6b = -6$$

$$\boxed{b=-1}$$

If $a=-3$ into x coefficient.

$$2(-3)b - (-3) = -9$$

$$-6b+3 = -9$$

$$-6b = -12$$

$$\boxed{b = 2}$$

$\therefore$ Two possible values of a and b.

$a = 3, \; b = -1$

$a = -3, \; b = 2.$

6a. $f^{-1}(-5,5) = \{x \in R \,/\, f(x) \in (-5,5)\}$

$= \{x \in R \,/\, -5 \leq f(x) \leq 5\}$

for $f(x) = 3x - 5. \; (x > 0)$

$-5 \leq 3x - 5 \leq 5.$

add 5 throughout.

$0 \leq 3x \leq 10 \quad \div 3.$

$0 \leq x \leq \frac{10}{3}.$

for $f(x) = \cancel{4}1 - 3x \; (x \leq 0)$

$-5 \leq 1 - 3x \leq 5.$

add -1 throughout

$-6 \leq -3x \leq 4 \quad \div$ by 3.

$-2 \leq -x \leq \frac{4}{3}.$

xPly by -1

$2 \geq x \geq -\frac{4}{3}.$

i.e $\frac{-4}{3} \leq x \leq 2.$

Combining $f^{-1}(-5,5) = \{x \in R \,/\, -\frac{4}{3} \leq x \leq 2$ or $0 \leq x \leq 10/3\}$

$= \{x \in R \,/\, \frac{-4}{3} \leq x \leq \frac{10}{3}\}.$

$= \left[\frac{-4}{3}, \frac{10}{3}\right].$

$f^{-1}[-6,5] = \{x \in R / f(x) \in [-6,5]\}$

$= \{x \in R / -6 \le f(x) \le 5\}$

for $f(x) = 3x-5$ $(x>0)$

$-6 \le 3x-5 \le 5$.

add 5

$-1 \le 3x \le 10$ $\div$ by 3.

$\frac{-1}{3} \le x \le \frac{10}{3}$.

for $f(x) = 1-3x$ $(x \le 0)$

$-6 \le 1-3x \le 5$.

add -1

$-7 \le -3x \le 4$. $\div$ by 3.

$-7/3 \le -x \le 4/3$.

xply by $= -1$

$7/3 \cancel{\ge} x \cancel{\ge} \frac{-4}{3}$.

i.e $\frac{-4}{3} \le x \le \frac{7}{3}$

Combining $f^{-1}[-6,5] = \{x \in R / \frac{-4}{3} \le x \le 7/3$ or $\frac{-1}{3} \le x \le \frac{10}{3}\}$

$= \{x \in R / \frac{-4}{3} \le x \le \frac{10}{3}\}$

$= \left[\frac{-4}{3}, \frac{10}{3}\right]$

b. i) for any. $a \in N$ or $b \in N$, we have $a-b=0$ is a divible by 5.

$\because$ 0 is divisible by any non zero integer. $(0 = 5 \times 0)$

$a-a$ is divisible by 5.

$\therefore$ $(a,a) \in R$ is reflexive.

ii) for any $a, b \in N$:

if $(a,b) \in N$ then $(b,a) \in R$.

By defn $a-b = 5k$ for some integer k.

Multiply both side by -1

$-(a-b) = -5k$.

$b-a = 5(-k)$.

Since $-k$ is integer, $b-a$ is divisible by 5

$\therefore (b,a) \in R$ is symmetric.

iii) for any $(a,b) \in N$ & $(b,c) \in N$ then $(a,c) \in N$.

Assume $(a,b) \in N$ & $(b,c) \in N$.

$a-b = 5k$ and $b-c = 5m$ for some integers k & m

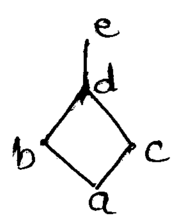
$(a-b) + (b-c) = 5k + 5m$.

$$(a-c) = 5(k+m).$$

Since $(k+m)$ is an integer, $a-c$ is divisible by 5.

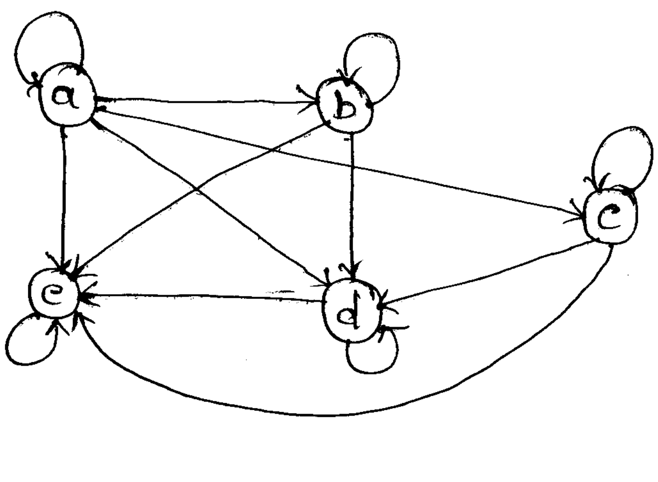
$\therefore (a,c) \in R$ is transitive.

c. $R = \left\{ \begin{array}{l} (a,a)\ (a,b)\ (a,c)\ (a,d)\ (a,e)\ (b,b) \\ (b,d)\ (b,e)\ (c,c)\ (c,d)\ (c,e)\ (d,d) \\ (d,e)\ (e,e) \end{array} \right\}$



i) $M(R) = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$

ii)



7a) Let S be the integer from 1 to 250

A : set of integers in S divisible by 3

B : set of integers in S divisible by 5

C : set of integers in S divisible by 7

* Number of integers divisible by 3

$|A| = \left\lfloor \frac{250}{3} \right\rfloor = 83$ —— 1m

* Number of integers divisible by 3 and 5

$(|A \cap B|) = \left\lfloor \frac{250}{15} \right\rfloor = 16$ —— 1m

* Number of integers divisible by 3 and 7

$(|A \cap C|) = \left\lfloor \frac{250}{21} \right\rfloor = 11$ —— 1m

* No. of integers divisible by 3, 5 and 7

$|A \cap B \cap C| = \left\lfloor \frac{250}{105} \right\rfloor = 2$ —— 1m

Using Inclusion Exclusion formula

$$|A \cap B^c \cap C^c| = |A| - |A \cap B| - |A \cap C| + |A \cap B \cap C|$$

$$= 83 - 16 - 11 + 2 = 85 - 37$$

$$= 58$$ # —— 2m

6m

7b) $F_{n+2} = F_{n+1} + F_n$, $F_0 = 0$ $F_1 = 1$

sol: $F_n = F_{n-1} + F_{n-2}$

A.E $k^2 - k - 1 = 0$

$k = \frac{1 \pm \sqrt{5}}{2}$

$k_1 = \frac{1+\sqrt{5}}{2}$ $k_2 = \frac{1-\sqrt{5}}{2}$

$F_n = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$ —— (2 m)

Put $n = 0, 1$

$\Rightarrow c_1 + c_2 = 0$ $\Rightarrow c_1 = -c_2$ —— (1m)

$\Rightarrow 1 = c_1\left(\frac{1+\sqrt{5}}{2}\right) + c_2\left(\frac{1-\sqrt{5}}{2}\right)$

$1 = \frac{1}{2}(c_1 + c_2) + \frac{\sqrt{5}}{2}(c_1 - c_2)$

$1 = 0 + \frac{\sqrt{5}}{2}(c_1 - c_2)$ Put $c_1 = -c_2$

$\Rightarrow c_2 = -\frac{1}{\sqrt{5}}$ $\Rightarrow c_1 = \frac{1}{\sqrt{5}}$ —— 3m

$F_n = \frac{1}{\sqrt{5}}\left[\frac{1+\sqrt{5}}{2}\right]^n - \frac{1}{\sqrt{5}}\left[\frac{1+\sqrt{5}}{2}\right]^n$ —— 1m

7m

7.c) Derangement is a permutation of things such that none of the things is in its original or ordered place. Numbers of possible derangements of n-distinct things 1, 2, -- n is denoted by D_n or d_n — (2m)

No of derangement of 1, 2, 3, 4 is 4! (1m)

$$D_n = n!\left\{1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \cdots + \frac{(-1)^n}{n!}\right\} \rightarrow (1m)$$

$n = 4$

$$D_4 = 4!\left\{1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right\}$$

$$= 24\left\{\cancel{1} - \cancel{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right\}$$

$$= 9$$ — (3m)

8a) C :

1	2
3	4

$n = r_1 = 4$

$r_2 = 2$

$r(C, x) = 1 + r_1 x + r_2 x^2$

$r(C, x) = 1 + 4x + 2x^2.$

8b) $a_n = 5a_{n-1} + 6a_{n-2}$ $\quad a_0 = 1 \quad a_1 = 3$

$a_n - 5a_{n-1} - 6a_{n-2} = 0$

$k^2 - 5k - 6 = 0$

$(k+1)(k-6) = 0 \quad k = -1, 6$

$\Rightarrow a_n = c_1(6)^n + c_2(-1)^n$

Put $n = 0$, $\quad 1 = c_1 + c_2$ —— (1)

Put $n = 1$ $\quad 3 = 6c_1 - c_2$ —— (2)

From (1) and (2)

$$\begin{array}{l} c_1 + \cancel{c_2} = 1 \\ 6c_1 - \cancel{c_2} = 3 \\ \hline 7c_1 \quad = 4 \end{array} \quad \Rightarrow c_1 = \frac{4}{7}$$

$c_2 = 1 - c_1 = 1 - \frac{4}{7} = \frac{3}{7}$

$\therefore a_n = \frac{4}{7}(6)^n + \frac{3}{7}(-1)^n$

8c) Multiple of 15 $(|A|) = \left\lfloor \frac{1000}{15} \right\rfloor = 66$

Multiple of 40 $(|B|) = \left\lfloor \frac{1000}{40} \right\rfloor = 25$

Multiple of 35 $(|C|) = \left\lfloor \frac{1000}{35} \right\rfloor = 28$

Multiple of 15 and 40 $(|A \cap B|) = \left\lfloor \frac{1000}{120} \right\rfloor = 8$

Multiple of 15 and 35 $(|A \cap C|) = \left\lfloor \frac{1000}{105} \right\rfloor = 9$

Multiple of 40 and 35 $(|B \cap C|) = \left\lfloor \frac{1000}{280} \right\rfloor = 3$

Multiples of 15, 40 and 35 $(|A \cap B \cap C|) = 1$

By Inclusion and Exclusion formula

$|A \cup B \cup C| = |A| + |B| + |C| - \{|A \cap B| + |A \cap C| + |B \cap C|\}$
$+ |A \cap B \cap C|$

$= (66 + 25 + 28) - \{8 + 9 + 3\} + 1$

$= 119 - 20 + 1$

$= 120 - 20$

$= 100$

9a) Let G be a non empty set and $*$ be a binary operation on G. Then G is called a group under the operation $*$, if the following conditions hold.

(1) $a * b \in G \;\; \forall \; a, b \in G$ [Closed under $*$]

(2) $(a * b) * c = a * (b * c)$ [$*$ is associative]

(3) Then $\exists$ an element $e \in G$ such that $a * e = e * a = a$ $\forall a \in G$ (Here 'e' is called identity Element)

(4) For every $a \in G$ there is an element $a' \in G$ such that $a * a' = a' * a = e$
(Here a' is called an inverse of a under $*$)

Eg: $(Z, +)$ is a group
$Z \rightarrow$ set of all integers.

* A nonempty subset H of a group G is called a subgroup of G whenever H itself is a group under binary operation in G

Eg: Under usual addition, the set of all even integers is a subgroup of the group of all integers.

9b) <u>Lagrange's theorem</u>

stmt: If G is a finite group and H is a subgroup of G, then the order of H divides order of G.

Since G is a finite group, H is finite

∴ No of cosets of H in G is finite.

let $Ha_1, Ha_2 \cdots Ha_r$ be the distinct right cosets of H in G. Then by the right coset decomposition of G, we have

$G = Ha_1 \cup Ha_2 \cup \cdots \cup Ha_r$ so that

$o(G) = o(Ha_1) + o(Ha_2) + \cdots + o(Ha_r)$

But $o(Ha_1) = o(Ha_2) = \cdots = o(Ha_r) = o(H)$

$\therefore o(G) = o(H) + o(H) + \cdots + o(H)$ (r times)

$o(G) = r\, o(H)$

$\Rightarrow o(H)$ divides $o(G)$

Hence so 12.

9c) ~~Kelsin~~ Klein 4 group is an abelian group with 4 Elements in which each element is self inverse and composing any two of the three non identity element produces the third one and is denoted by K_4

$A = \{e, a, b, c\}$

$a^2 = b^2 = c^2 = e \quad a \circ b = c \quad b \cdot c = a \quad c \cdot a = b$

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

closure property holds

$\because a, b \in A \Rightarrow a * b = c \in A.$

A is an abelian group under multiplication

$\because a * b = b * a$

$c = c \quad \forall a, b \in A$

e' is identity Element

Every Elements is its own inverse

$\therefore A = \{e, a, b, c\}$ is klein 4 group

10a) Let G be group and H & K be subgroups of G. Take $a, b \in H \cap K$. Then $a, b \in H$ & $a, b \in K$. Therefore $ab^{-1} \in H$ and $ab^{-1} \in K$, because H and K are subgroups of G
$\Rightarrow ab^{-1} \in H \cap K$. Hence $H \cap K$ is a subgroup of G.
Thus intersection of two subgroups of a group G is a subgroup of G.

10b) Let $G = \{1, w, w^2\}$. where $w^3 = 1$

$*$	1	w	w^2
1	1	w	w^2
w	w	w^2	1
w^2	w^2	1	w

* Closure property
Take any two elements in G. Their product is also a cube root of unity
ie $1 \cdot 1 = 1$, $1 \cdot w = w$, $1 \cdot w^2 = w^2$ etc

* Associative property
$(ab)c = a(bc)$
From table $1, w, w^2 \in G$.
$(1 \cdot w) w^2 = 1 (w \cdot w^2) = 1$.
Associative property holds

* Identity Element : $1 \in G$

$\Rightarrow 1 \cdot w = w \cdot 1 = w \in G \quad \forall\, w \in G.$

* Inverse Element :

Inverse of 1 is 1

Inverse of w is w^2 $\because w \cdot w^2 = w^3 = 1$

Inverse of w^2 is w.

since $G = \{1, w, w^2\}$ satisfies closure associativity, identity and inverse

$\Rightarrow$ cube root of unity form a group under multiplication

10c) let $G = S_4$

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$$

since $G = S_4 \Rightarrow |G| = 4! = 24$

To find subgroup of $H = \langle \alpha \rangle$

$\alpha^1 = (1\ 2\ 3\ 4)$

$\alpha^2 = (1\ 3)(2\ 4)$

$\alpha^3 = (1\ 4\ 3\ 2)$

$\alpha^4 = e$

since $\alpha^4 = e$ the order of α is 4

$\therefore$ subgroup generated by α is

$$H = \langle \alpha \rangle = \{e, \alpha, \alpha^2, \alpha^3\}$$

$\Rightarrow |H| = 4$

By Lagrange's theorem, the number of left cosets of H in $G = \frac{|G|}{|H|}$

$$= \frac{24}{4} = 6$$

3/2/25
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